

1. Create a dataset of 50 random integers between 10 and 100. Calculate mean, median, and standard deviation using Python.
2. Generate two random numerical arrays and compute:
  - Covariance
  - Pearson correlation coefficient
  - Spearman correlation coefficient
3. Create a skewed dataset and calculate its skewness using Python.
4. Generate a normal distribution dataset and calculate kurtosis.
6. Simulate a normal distribution with mean = 50 and std = 5. Plot the distribution.
7. Generate a binomial distribution ( $n=10$ ,  $p=0.5$ ). Visualize using a bar chart.
8. Generate a Poisson distribution ( $\lambda=3$ ). Plot the probability mass function.
9. From a dataset of 1000 random values:
  - Take a random sample of 100
  - Take a systematic sample
  - Take a stratified sample
10. Demonstrate the Central Limit Theorem by:
  - Taking 50 different samples
  - Plotting the sampling distribution of the mean
11. Perform a one-sample t-test on a dataset to check if the population mean equals a given value.
12. Perform an independent two-sample t-test between two datasets.
13. Perform a chi-square test for independence on a categorical dataset.
14. Calculate and interpret a 95% confidence interval for the mean.
15. Create a simple linear regression model using `statsmodels`.
16. Interpret regression output: R-squared, p-value, coefficients.
17. Perform multiple linear regression with at least two independent variables.
18. Generate time-series data and perform trend analysis using linear regression.
19. Plot regression line over scatter plot.
20. Write a Python function that:
  - Loads a dataset
  - Displays summary statistics
  - Plots histogram for all numerical columns